Development and validation of a unified OSATS tool for open surgical knot-tying.

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Abstract

BACKGROUND: Proficiency in surgical knot tying is a fundamental skill for surgical trainees, yet objective assessment tools are limited. PURPOSE: This study aimed to develop and provide initial reliability and validity evidence for a unified Objective Structured Assessment of Technical Skills (OSATS) tool tailored for open surgical knot-tying across multiple knot types. METHODS: A prospective observational study was conducted with fifty medical students at Vajira Hospital. All participants received structured simulation-based training using the Modified Peyton's Four-Step Approach, covering Square, Surgeon's, Aberdeen, and Roeder knots. Each participant performed three repetitions of each knot on a synthetic tissue model. Performances were video-recorded and assessed by two experienced surgeons using a newly developed OSATS tool consisting of a General Technical Skills (GTS) checklist and a Global Rating Scale (GRS). Reliability and validity were statistically analysed. RESULTS: The unified OSATS tool demonstrated inter- and intra-rater reliability. Mean global rating scores did not significantly differ across knot types ($p = 0.217$), providing initial internal structure evidence across knot types. Student feedback indicated high acceptability and perceived learning confidence following the structured training programme. CONCLUSION: This unified OSATS tool provides a reliable and practical framework for assessing open surgical knot-tying skills. As implemented within a structured training programme, the tool appears feasible for formative assessment in novice learners. Broader application with additional raters, objective performance metrics (e.g., time and errors), and multi-institutional samples is recommended to strengthen validity evidence and generalisability.

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